

1 Basic operations with vectors and matrices

- Scalars can be summed with scalars, vectors with other vectors of the same size, and matrices with matrices of the same size. Therefore, $u+v$ and $B+C$ are mathematically well-defined. MATLAB interprets $s + A$ by adding s to each component of A . This is not the sum of a scalar and matrix in a well-defined mathematical sense. Instead,

MATLAB implicitly interprets s as $\begin{pmatrix} 2 & 2 \\ 2 & 2 \\ 2 & 2 \end{pmatrix}$ and adds it to A as a matrix. You can see this explicitly by evaluating $s + A - A$ with MATLAB.

- Vectors can be combined with an inner product if they have the same dimensionality. This is true for u and v , and their inner product is $u \cdot v = 0 * 2 + 1 * 5 = 5$. Note that I used the notation $\cdot$ to denote the inner product, and inner products are often referred to as “dot products.” I used the notation $*$ to denote the usual multiplication of numbers. I will use these notations throughout the course.
- The norm of a vector is the length of the vector. The norm of a vector is the square root of vector’s inner product with itself.
- Any two vectors can be combined to produce a matrix through an outer product. If the first vector is N -dimensional and the second vector is M -dimensional, then the resulting matrix is $N \times M$. Therefore, the possible pairs are $\{u, v\}$, $\{u, w\}$ and $\{v, w\}$, and one can produce six distinct mathematical objects by taking the outer product of u with v , v with u , u with w , and so on.
- A matrix can multiply a vector when its number of columns matches the dimensionality of the vector. The dimensionality of the resulting vector corresponds to the number of rows of the matrix. Here any of the matrices can multiply u or v , and none of them can multiply w . For example,

$$Av = \begin{pmatrix} 0 & 0.5 \\ 10 & 1 \\ 2 & 9 \\ 2 & 7 \end{pmatrix} \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 * 2 + 0.5 * 5 \\ 10 * 2 + 1 * 5 \\ 2 * 2 + 9 * 5 \\ 2 * 2 + 7 * 5 \end{pmatrix} = \begin{pmatrix} 2.5 \\ 25 \\ 49 \\ 39 \end{pmatrix}. \quad (1)$$

- Two matrices can be multiplied together when the number of columns in the first matrix matches the number of rows in the second matrix. If the first matrix is $N \times K$ and the second matrix is $K \times M$, then the resulting matrix is $N \times M$. Here the valid

matrix products are AB , AC , BC and CB . For example,

$$BC = \begin{pmatrix} 4 & 4 \\ 8 & 9 \end{pmatrix} \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix} = \begin{pmatrix} 4*3 + 4*6 & 4*4 + 4*8 \\ 8*3 + 9*6 & 8*4 + 9*8 \end{pmatrix} = \begin{pmatrix} 36 & 48 \\ 78 & 104 \end{pmatrix}. \quad (2)$$

Note that $BC \neq CB$, and one generally has to be careful about how matrices are ordered in equations.

- An identity matrix is a square matrix that multiplies vectors or matrices without changing them. It always has ones along the diagonal and zeros elsewhere. The 3×3 identity matrix is $I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.
- The matrix transpose swaps the rows and columns of a matrix. So, $A_{ij}^T = A_{ji}$ and

$$A^T = \begin{pmatrix} 0 & 10 & 2 & 2 \\ 0.5 & 1 & 9 & 7 \end{pmatrix}. \quad (3)$$

Since A^T is a 2×4 matrix, it can multiple the 4-dimensional vector w .

- We may express the norm of w as $\|w\| = \sqrt{w^T w}$.
- The matrix inverse inverse is only defined for a subset of square matrices. When it exists, it is defined as the unique square matrix that results in the identity matrix when multiplied with the original matrix on either side. A is not invertible because it is not square. Evaluation in MATLAB shows that

$$B^{-1} = \begin{pmatrix} 9/4 & -1 \\ -2 & 1 \end{pmatrix}. \quad (4)$$

Note that $BB^{-1} = B^{-1}B = I$. Trying to invert C in MATLAB produces a warning that “the matrix is singular to working precision.” This means that MATLAB was unable to find a matrix inverse, but the program is unsure whether a more accurate numerical algorithms could produce one. Rather than crash, MATLAB returns the nonsense answer that $C^{-1} = \begin{pmatrix} \infty & \infty \\ \infty & \infty \end{pmatrix}$.

- The determinant is defined for square matrices, so it can be computed for s , B , and C . For a 1×1 matrix (i.e., a scalar), the determinant leaves the scalar unchanged. Thus, $\det s = 2$. For 2×2 matrices, the determinant is the product of the diagonal elements minus the product of the off-diagonal elements. Thus, $\det B = 4*9 - 4*8 = 4$ and $\det C = 3*8 - 4*6 = 0$. Determinants of general square matrices can be evaluated by hand using the Leibniz formula or the Laplace expansion, but this is tedious and numerical methods are usually applied to evaluate the determinants of square matrices larger than 3×3 . A square matrix is invertible if and only if its determinant is nonzero.
- Hadamard products can be used to combine the vectors u and v or the matrices B and C .

2 Random processes and probability

2.1 Bernoulli distribution

The probability mass function specifies the probability of observing each possible value. Here it is

$$P(k) = \begin{cases} 1-p & , \text{ if } k = 0 \\ p & , \text{ if } k = 1 \end{cases} . \quad (5)$$

The cumulative density function is defined to be the probability that the value is less than or equal to any real number. Here it is

$$F(x) = P(k \leq x) = \begin{cases} 0 & , \text{ if } x < 0 \\ 1-p & , \text{ if } 0 \leq x < 1 \\ 1 & , \text{ if } x \geq 1 \end{cases} . \quad (6)$$

The mean of any discrete probability distribution is computed by $\mu = \sum_k P(k)k$. Here it is

$$\mu = (1-p) * 0 + p * 1 = p. \quad (7)$$

The variance of any discrete probability distribution is computed by $\sigma^2 = \sum_k P(k)(k - \mu)^2$. Here it is

$$\mu = (1-p) * (0 - p)^2 + p * (1 - p)^2 = p(1-p)(p + 1 - p) = p(1-p). \quad (8)$$

The standard deviation is the square root of the variance. Here it is $\sqrt{p(1-p)}$.

2.2 Exponential distribution

For continuous probability distributions, we compute the cumulative density function by integrating the probability density from $-\infty$ upwards.

$$F(x) = \int_{-\infty}^x dx' P(x'). \quad (9)$$

Here it is zero for all negative values of x . For non-negative values, it is

$$x \geq 0 \implies F(x) = \int_0^x dx' \lambda e^{-\lambda x'} = -e^{-\lambda x'} \Big|_{x'=0}^{x'=x} = 1 - e^{-\lambda x}. \quad (10)$$

Putting these two pieces together, we have

$$F(x) = \begin{cases} 0 & , \text{ if } x < 0 \\ 1 - e^{-\lambda x} & , \text{ if } x \geq 0 \end{cases} . \quad (11)$$

The probability that the value is in (a, b) is

$$P(a < x < b) = F(b) - F(a) = e^{-\lambda a} - e^{-\lambda b}. \quad (12)$$

Note that this probability is zero if $a = b$. The mean of a continuous probability distribution is computed by integration, $\mu = \int_{-\infty}^{\infty} P(x)x$. Here it is

$$\mu = \int_0^{\infty} dx \lambda e^{-\lambda x} x = -x e^{-\lambda x} \Big|_{x=0}^{x=\infty} - \int_0^{\infty} dx (-e^{-\lambda x}) = -\frac{1}{\lambda} e^{-\lambda x} \Big|_0^{\infty} = \frac{1}{\lambda}, \quad (13)$$

where we used the integration by parts rule from calculus. The median is the value for which the cumulative density function is equal to 0.5. Here it is defined by

$$0.5 = 1 - e^{-x_{1/2}\lambda} \implies -x_{1/2}\lambda = \ln(0.5) \implies x_{1/2} = -\frac{\ln(0.5)}{\lambda} = \frac{\ln(2)}{\lambda}, \quad (14)$$

where $x_{1/2}$ denotes the median and $\ln$ is the natural logarithm. The mode is the value that maximizes the probability density function. Here it is 0. The standard deviation is the square root of the variance, and the variance of a continuous distribution is computed by integration $\sigma^2 = \int_{-\infty}^{\infty} P(x)(x - \mu)^2 = -\mu^2 + \int_{-\infty}^{\infty} P(x)x^2$. Here we can again use integration by parts to compute this integral

$$\int_0^{\infty} dx \lambda e^{-\lambda x} x^2 = -x^2 e^{-\lambda x} \Big|_{x=0}^{x=\infty} - 2 \int_0^{\infty} dx (-e^{-\lambda x}) x = \frac{2}{\lambda} \int_0^{\infty} dx \lambda e^{-\lambda x} x = \frac{2}{\lambda^2}. \quad (15)$$

We therefore find

$$\sigma = \sqrt{\sigma^2} = \sqrt{-\frac{1}{\lambda^2} + \frac{2}{\lambda^2}} = \sqrt{\frac{1}{\lambda^2}} = \frac{1}{\lambda}. \quad (16)$$

Note that the standard deviation is equal to the mean for the exponential distribution.

3 Multivariate probability distributions

3.1 Bivariate normal distribution

The means of x_1 and x_2 are μ_1 and μ_2 . They describe the expected (or average) values. The variances of x_1 and x_2 are σ_1^2 and σ_2^2 . These describe the spread around the means. The correlation coefficient between x_1 and x_2 is c . This parameter describes the extent to which fluctuations of x_1 away from μ_1 are accompanied by fluctuations of x_2 away from μ_2 .

3.2 Marginal distributions

All marginal distributions of a multivariate normal distribution are again normal. As suggested by the notation and meaning of the parameters, the variable x_1 will be normally distributed with mean μ_1 and variance σ_1^2 .

3.3 Statistical independence

Two normally distributed variables are statistically independent if and only if they are uncorrelated, which means $c = 0$. More generally, statistically independent variables are always uncorrelated, but uncorrelated variables need not be statistically independent.

3.4 Conditional distributions

If x_1 and x_2 are statistically independent, then

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} = \frac{P(x_1)P(x_2)}{P(x_2)} = P(x_1). \quad (17)$$

For a bivariate Gaussian with the specified parameters, the mean of $P(x_1|x_2 = 0.75)$ is smaller than μ_1 . This is because x_1 and x_2 are positively correlated and $0.75 < \mu_2$.

3.5 Bayes rule

By definition,

$$P(x_2|x_1) = \frac{P(x_1, x_2)}{P(x_1)}. \quad (18)$$

Also by definition,

$$P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)} \implies P(x_1, x_2) = P(x_1|x_2)P(x_2). \quad (19)$$

The latter expression is called the product rule of probability. Substituting Eq. (19) into Eq. (18), we find,

$$P(x_2|x_1) = \frac{P(x_1|x_2)P(x_2)}{P(x_1)}. \quad (20)$$

This is Bayes rule. In Bayesian terminology, we say that the posterior distribution (i.e., $P(x_2|x_1)$) is proportional to the product of the likelihood function (i.e., $P(x_1|x_2)$) times the prior probability of x_2 (i.e., $P(x_2)$). The term $P(x_1)$ normalizes the distribution.